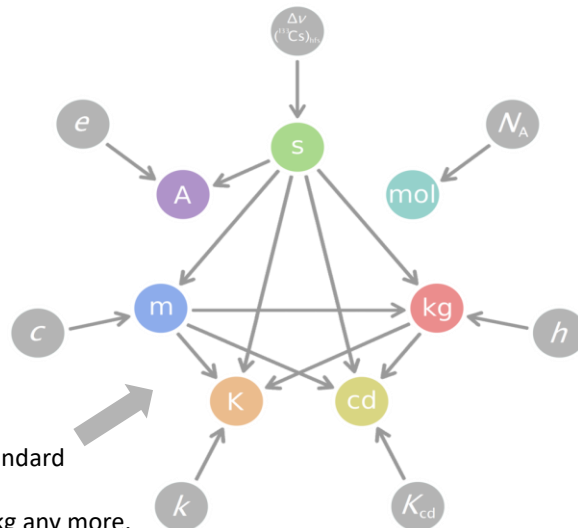


Relations of the quantities



- No more etalons, like the Pt mass standard
- Kilogram depends on m and s .
- Ampere does not depend on m and kg any more.
- Mol does not depend on kg any more.
- Kelvin depends on m , kg and s .

Measurement methods and related physical phenomena

Already discussed before

6

Time standards, atomic clocks

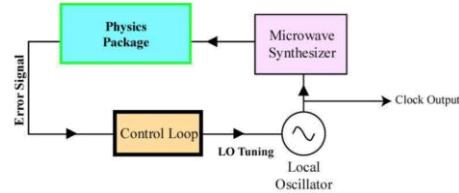
Principle of operation:

- We have a physical system with a very sharp resonance, i.e. $Q = \nu_0 / \gamma$ is large, where ν_0 is the resonant frequency, γ is the resonance width.

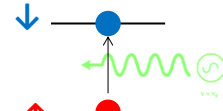
- In the case of atomic clocks, the physical system is two well-defined (hyperfine) states of the atom, whose energy difference is $E = h \cdot \nu_0$. If an EM wave of frequency corresponding to the energy difference is radiated, the transition between the two states can be excited resonantly.

- Local oscillator (e.g. quartz) from which the clock operates. Its timing is controlled by a feedback loop: a signal close to ν_0 is synthesized from the local oscillator signal (see high frequency clocks) and sent to the physical system. The response of the system determines whether the synthesized frequency is equal to ν_0 . If not, the local oscillator's frequency is tuned.

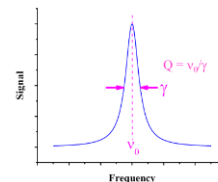
Synchronization principle with a high quality factor (Q) physical system



Excitation of a two-state system with EM wave, $E_{\uparrow} - E_{\downarrow} = h\nu_0$



Quality factor of a resonance curve



Source: Robert Lutwak: Principles of Atomic Clocks EFTF –IFCS 2011

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Time standards, atomic clocks

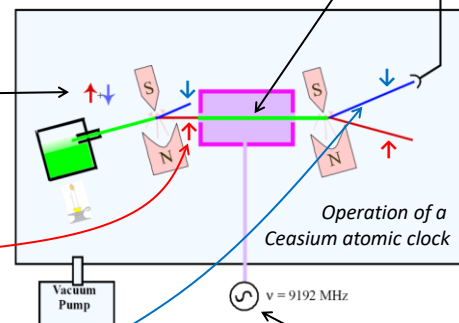
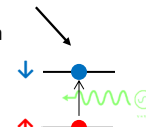
The interaction of the electron's magnetic moment with the nuclear magnetic moment yields a so-called hyperfine splitting of the ground-state energy level of the ^{133}Cs atoms. The energy difference between these two hyperfine states is $E = h \cdot \nu_0$, where $\nu_0 = 9,192,631,770 \text{ 1/s}$. **This frequency serves as the time standard.** The electron spin (or the electron's magnetic moment) is opposite in the two states. Once the ^{133}Cs atom in the ground state is irradiated by an EM wave of the same ν_0 frequency a transition occurs to the excited state. Meanwhile, the electron's magnetic moment is reversed, the electron's magnetic moment is reversed, a key element in detection.

Detection is like the Stern-Gerlach experiment (1922). A source emits ^{133}Cs atoms. In this atom beam the ground state magnetic moment can either be $\uparrow$ or $\downarrow$.

A μ magnetic moment moving in a magnetic field gradient feels a force of $F = \text{grad}(\mu \cdot \mathbf{B})$. This force is opposite for the $\uparrow$ and $\downarrow$ states, i.e. an inhomogeneous magnetic field deflects the atoms with $\uparrow$ and $\downarrow$ ground states in different directions. This way we select the atoms with $\uparrow$ ground state (red beam).

This red beam is irradiated with a microwave of ν frequency, with the goal of exciting many atoms to the $\downarrow$ state. With another Stern Gerlach experiment (inhomogeneous B field) we again separate the $\uparrow$ and states $\downarrow$ to two different directions and measure the atom intensity in the $\downarrow$ state.

Excitation of the spin down electron in the ground state to a spin up excited state by EM irradiation



Finally, we tune ν such that the the largest $\downarrow$ intensity is detected. In this case ν matches ν_0 , i.e. the excitation frequency serves as the standard.

Source: Robert Lutwak: Principles of Atomic Clocks EFTF –IFCS 2011

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Time standards, atomic clocks

Nowadays the best atomic clocks reach an accuracy of $<10^{-15}$ s (fountain atomic clocks with ultracold atoms, NIST-F2)

2011: battery-powered pocket atomic clocks: $<10^{-10}$ s accuracy.

Overview

Applications

Documents

An unmatched combination of breakthroughs - reduced Size, Weight and Power consumption (SWaP) - brings the accuracy and stability of atomic clocks to portable applications.

The Microsemi Quantum™ SA.45s CSAC is the world's first commercially available chip scale atomic clock, providing the accuracy and stability of atomic clock technology while achieving true breakthroughs in reduced size, weight and power consumption.

Overview

Atomic clock accuracy and low power make the SA.45s ideal for portable applications requiring precise synchronization and time keeping, especially in GPS-denied environments.

Breakthrough Leadership

The CSAC's key specs include:

- $<120\text{mW}$ power consumption
- $<17\text{cm}^3$ volume
- 35g weight
- $\pm 5.0\text{E}-11$ accuracy at shipment
- $<5\text{E}-12$ at $\tau = 1$ hour short-term stability (Allan Deviation)
- $<3.0\text{E}-10/\text{month}$ aging rate



Quantum™ SA.45s Chip Scale Atomic Clock

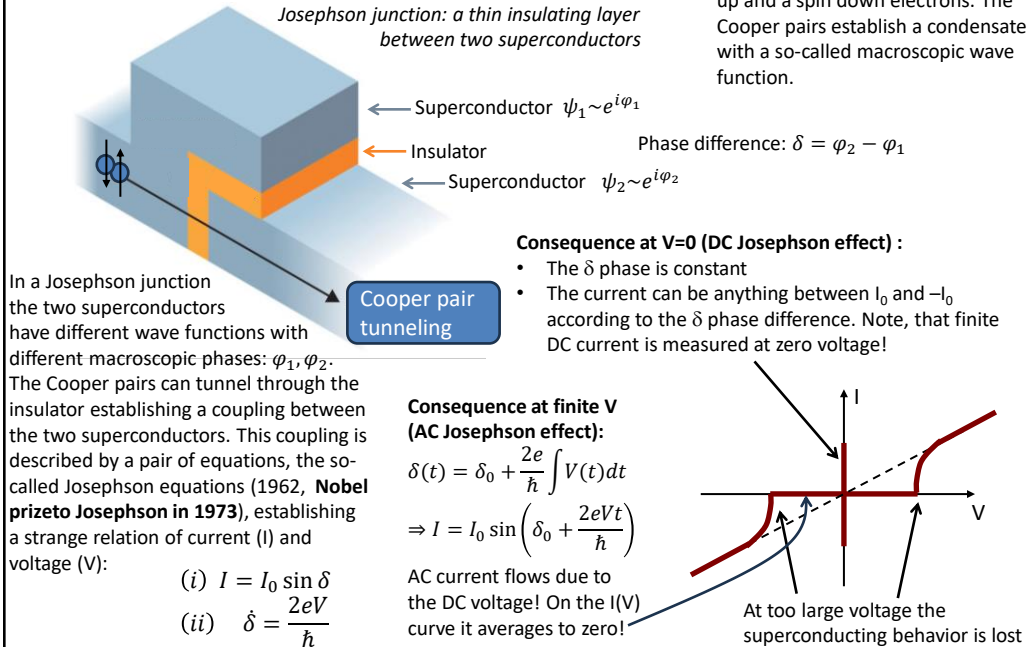
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Measurement methods and related physical phenomena

	Definition	Measurement principles, standards and related quantum phenomena
s	Fixing the hyperfine splitting of the Caesium 133 atom, $\Delta\nu(^{133}\text{Cs})$.	^{133}Cs atomic clocks ✓
K	Fixing the Boltzmann constant (k).	Speed of sound in noble gases, Johnson-Nyquist noise
m	Fixing the speed of light (c).	Laser light interference ✓
mol	Fixing the Avogadro number (N_A).	-
A	Fixing the elementary charge (e).	Josephson effect + Quantized Hall effect (or electron pump?)
kg	Fixing the Planck constant (h).	Watt balance, Avogadro-project
cd	The luminous intensity of a monochromatic ($f=540\times 10^{12}$ Hz) source emitting 1/683W per unit solid angle.	...

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Converting voltage to time: the Josephson effect



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Converting voltage to time: the Josephson effect

→ How to detect the AC Josephson effect? Let us apply a microwave irradiation!

$$V(t) = V_0 + V_1 \cos \omega t \Rightarrow \delta(t) = \delta_0 + \frac{2e}{\hbar} \int V(t) dt = \delta_0 + \frac{2eV_0}{\hbar} t + \frac{2eV_1}{\hbar \omega} \sin \omega t$$

$$I = I_0 \sin \left(\delta_0 + \frac{2eV_0}{\hbar} t + \frac{2eV_1}{\hbar \omega} \sin \omega t \right) \approx \text{Taylor expansion}$$

$$I_0 \sin \left(\delta_0 + \frac{2eV_0}{\hbar} t \right) + I_0 \frac{2eV_1}{\hbar \omega} \sin(\omega t) \cos \left(\delta_0 + \frac{2eV_0}{\hbar} t \right) =$$

$$= I_0 \sin \left(\delta_0 + \frac{2eV_0}{\hbar} t \right) + I_0 \frac{2eV_1}{\hbar \omega} \left(\sin \left(\omega t - \frac{2eV_0}{\hbar} t - \delta_0 \right) + \sin \left(\omega t + \frac{2eV_0}{\hbar} t + \delta_0 \right) \right)$$

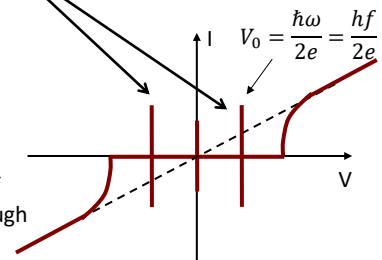
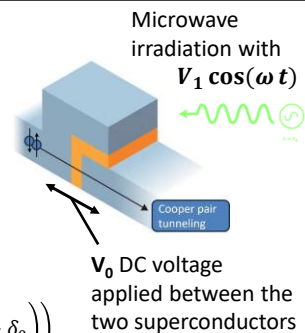
At $V_0 = \pm \hbar \omega / 2e$ the time-dependence cancels, i.e. DC current flows! This is called **Shapiro resonance**, which appears in the I(V) curve as shown in the figure:

Note, that this resembles the **heterodyne detection** and the appearance of the difference frequency.

Measuring the Shapiro resonances we can **convert the V_0 voltage into a frequency f based on two natural constants**, the Planck constant (h) and the electron charge (e):

$$V_0 = \frac{hf}{2e}$$

This is the principle of the **voltage standard**, i.e. the conversion of voltage measurement to highly accurate time measurement through the AC Josephson effect. Accuracy: 2×10^{-8} (NIST 2014).



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Classical Hall effect

Edwin Hall (1879). Electrons move in a thin sample applying a magnetic (B_z) field perpendicular to the sample plane. Current is injected in the x direction, yielding a v_e drift electron velocity opposite to the current. Due to the B field a Lorentz force acts on the electrons in the y direction $F_{L,y} = eB_z v_e$, deflecting them towards one side of the sample. Due to the charge separation, a V_H Hall voltage appears in the perpendicular direction yielding an $E_y = V_H/w$ electric field in the y direction. In equilibrium the electric force in the y direction balances the Lorentz force :

$$ev_e B_z = -eE_y = -eV_H/w$$

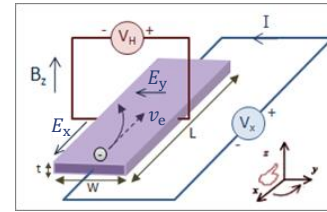
On the other hand, the current density is calculated from the n electron density as:

$$j_x = -nev_e \quad \text{From these a so-called Hall resistance can be calculated: } R_H = \frac{E_y}{j_x} = \frac{B_z}{ne}$$

$$\text{In a two-dimensional sample } I = j_x w, \text{ so: } R_H = \frac{E_y}{j_x} = \frac{V_H}{I} = \frac{B_z}{ne}$$

Hall resistance (R_H):

- Linearly increases with B
- Independent of the τ scattering time (see Drude model and the derivation of the thermal noise), only depends on the electron density: a **perfect method to determine the electron density** in semiconductors
- In p-type semiconductors holes are conducting instead of electrons, i.e. the Hall resistance has opposite sign.
- Widely applied in magnetic field sensors (see later)

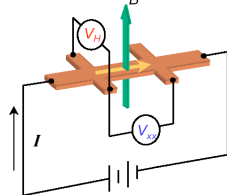


Source: Wikipedia

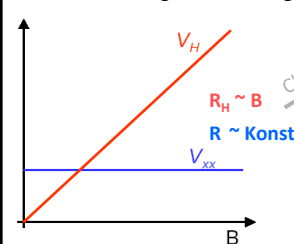
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Converting current to voltage through the Quantized Hall effect

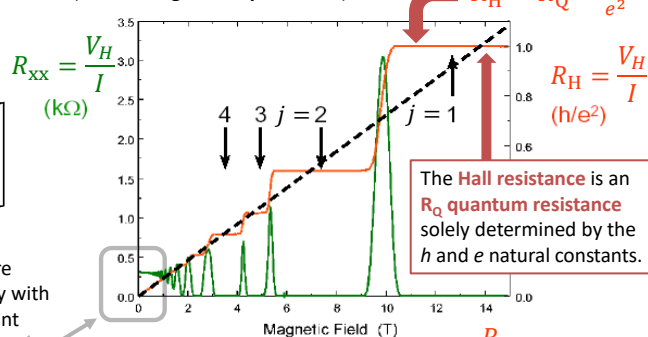
The **quantized Hall effect** is measured in a **macroscopic 2D semiconducting structures (two-dimensional electron gas, 2DEG)** at **low temperature in high magnetic field** using the so-called Hall-bar geometry:



The **Classical Hall effect** with yield the figure below, the V_H hall voltage increases linearly with B and the V_{xx} longitudinal voltage is constant



At high enough B, low enough T and clean enough 2DEG samples a completely different, **quantized behavior is observed** (von Klitzing, Nobel prize 1985):



$$R_{Hall} = \frac{h/e^2}{j} = 25812.807 \text{ Ohms / integer}$$

- R_Q is independent of the shape, size, material of the sample
- R_Q has an accuracy of $3 \cdot 10^{-10}$! (NIST 2014).
- R_Q accurately converts current to voltage (**resistance standard**)
- R_Q makes the I to V conversion, the AC Josephson effects makes the V to I conversion, the both are used to convert **Ampères to Seconds**

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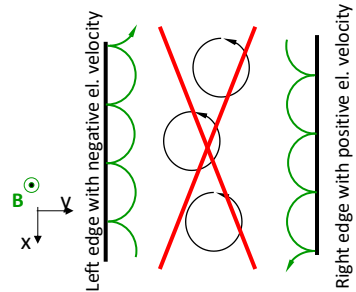
Some explanation of the Quantized Hall effect

Edge channels: the Lorentz force pushes the electrons with positive x velocity to the one side and the electrons with negative x velocity to the other side of the sample. All the current is carried by these so-called edge channels, there is no current at the centre of the sample.

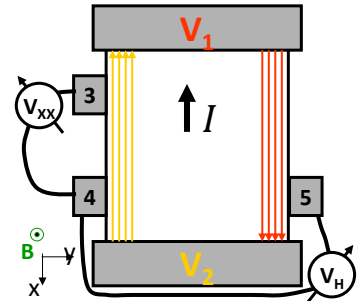
Zero longitudinal resistance: where the quantized Hall resistance is observed the V_{xx} longitudinal voltage drop is zero (see previous slide), which indicates perfect (resistanceless) conduction. This is related to the fact that the electrons coming from electrode #1 move on one side of the sample. To be reflected back to the same electrode they should have opposite x velocity, which is only possible at the other side of the sample. The electrons cannot scatter from one edge channel to the other because of the far distance of the two sides. This means, that an electron starting from electrode #1 will always arrive to electrode #2, it cannot be scattered back to #1.

Quantized energy levels: the finite magnetic field introduces quantized energy levels (Landau levels) due to the quantization of the cyclotron orbitals. If only a single quantized energy level is available, the conductance through the edge channel is $G = G_Q = \frac{e^2}{h}$. For n available energy levels it is $G = n \cdot \frac{e^2}{h}$, and n is varied by the strength of the magnetic field.

Classical representation of the electron Trajectories inside and at the edges



Hall geometry with 4 quantized edge channels



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Converting current directly to time: the electron pump

Electron pump: two tunneling barriers are separating a tiny confined region (**quantum dot**) from the two electrodes. Inside the quantum dot only zero or one electron can stay, having two electrons would cost too much (unavailable) Coulomb energy.

The potential barriers of the two tunnel junctions are **periodically modulated with f frequency** using the phase shifting scheme in the figure. It is then possible to pass a **single electron per period**. This yields a current:

$$I = e \cdot f$$

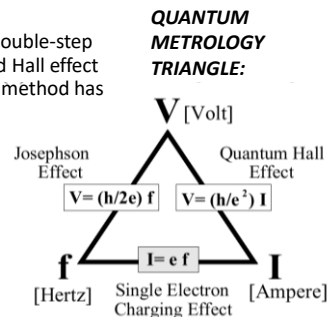
→ The electron pump yields a **direct conversion from current measurement to time measurement**. Accuracy: $\sim 10^{-7}$

In summary: today the **current (Ampere) standard** is still the double-step measurement of converting current to voltage by the quantized Hall effect and converting voltage to time by the Josephson effect, as this method has the best accuracy.

The electron pump offers a direct current to time conversion, though with somewhat worse accuracy.

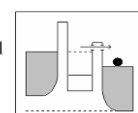
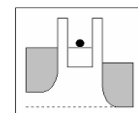
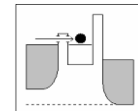
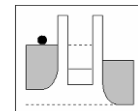
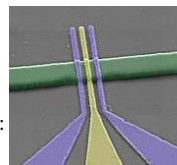
The mutual relation between time (s), voltage (V) and current (A) is described by the so-called **quantum metrology triangle**:

Note: in the $I \rightarrow V \rightarrow f$ conversion the **Planck constant cancels**, only **e remains**: $I \rightarrow (e^2/h)U \rightarrow (e^2/h) \cdot (h/2e)f = e \cdot f/2$



Operation principle of the electron pump

GaAs/AlGaAs quantum dot



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Kouwenhoven et al., Z. Phys. B **85**, 381 (1991) <http://arxiv.org/abs/1204.6500>

Measurement methods and related physical phenomena

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s	Fixing the hyperfine splitting of the Caesium 133 atom, $\Delta\nu(^{133}\text{Cs})$.	^{133}Cs atomic clocks ✓
K	Fixing the Boltzmann constant (k).	Speed of sound in noble gases, Johnson-Nyquist noise
m	Fixing the speed of light (c).	Laser light interference ✓
mol	Fixing the Avogadro number (N_A).	-
A	Fixing the elementary charge (e).	Josephson effect + Quantized Hall effect (or electron pump?) ✓
kg	Fixing the Planck constant (h).	Watt balance, Avogadro-project
cd	The luminous intensity of a monochromatic ($f=540 \times 10^{12}$ Hz) source emitting 1/683W per unit solid angle.	...

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Temperature measurement methods

Speed of sound in gas (argon)

Principle: If the gas is hotter, the atoms travel faster, and thereby the sound travels faster.

The speed of sound can be calculated in the ideal gas picture: $c_{ideal} = \sqrt{\frac{\gamma k T}{m}}$

For noble gases: $\gamma=5/3$. Accuracy $\sim 2 \times 10^{-6}$.

Difficulties: The gas temperature must be accurately kept, isotopic composition and volume must be known. Difficult to use as a standard over a wide range.

ITS-90 International Temperature Scale

It gives calibration points on how to measure T between 0.5K - 1350K, based on a numerous physical systems. E.g. Pt piece resistance, helium vapor pressure, phase transition points, e.g. triple point for oxygen (54.35K), freezing point for gold (1337K).

Johnson-Nyquist noise (thermal noise) thermometer

Principle: spectral density of the thermal noise of a resistor R :

$$S_V = 4kTR, \quad \Delta V_{\text{rms}} = \sqrt{4kT \cdot R \cdot \Delta f} \quad (\text{see thermal noise lecture})$$

Benefit: Background theory is exact. It works over a wide T range.

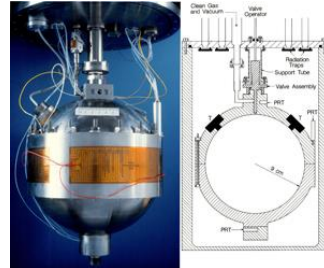
Difficulties: small signal to measure, other noise pickups. Cross-correlation technique is used (see noise lectures).

If wide T range is important then a reference signal is measured simultaneously (AC Josephson Quantized Voltage Noise Source, see Fig.) S unit alternates the two signals to be measured between unknown $R(T)$ and the reference signal.

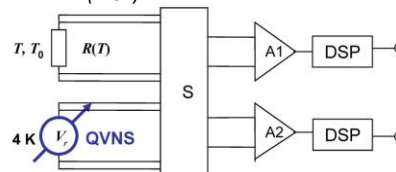
Slow drift of amplifiers can be compensated.

(Experience it next semester during the noise measurement!)

Instrument for sound velocity measurement in argon gas (NIST)



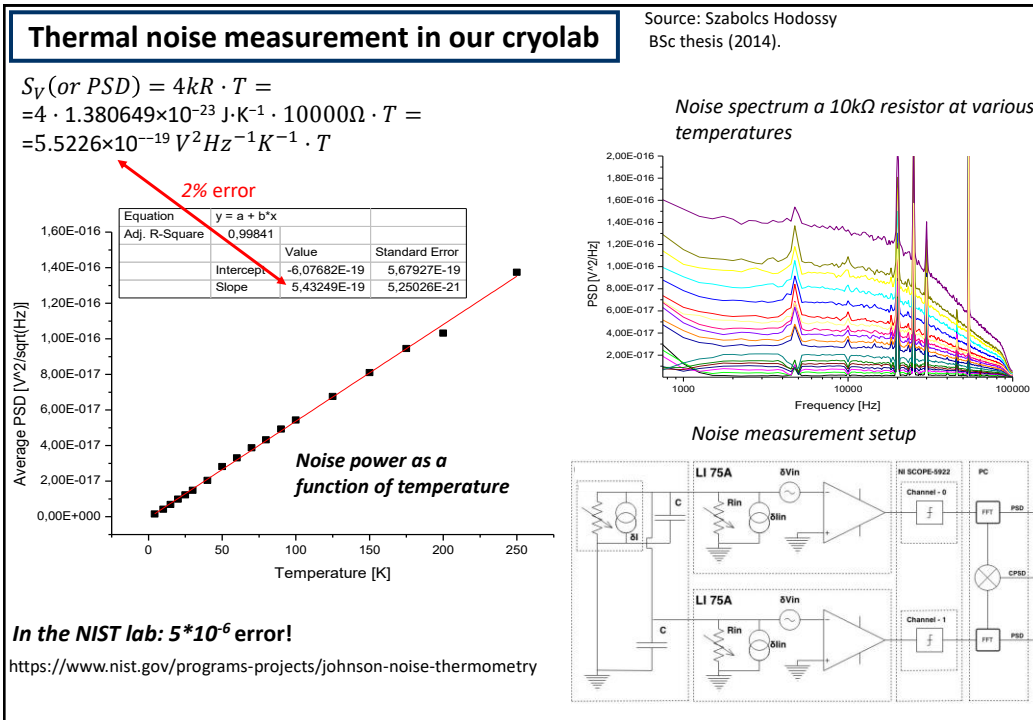
Johnson-Nyquist noise thermometer (NIST)



Source:

<https://www.nist.gov/programs-projects/johnson-noise-thermometry>
<https://www.measurement.govt.nz/metrology/si-units/kelvin/>

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Measurement methods and related physical phenomena		
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kg	Fixing the Planck constant (h).	Watt balance, Avogadro-project
cd	The luminous intensity of a monochromatic ($f=540 \times 10^{12}$ Hz) source emitting 1/683W per unit solid angle.	...

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Watt balance: converting mass to voltage, current and velocity measurement

Watt balance: A precision instrument for **measuring mass based on current and voltage by fixing the Planck constant (h)**.

Principle: A current-carrying conductor in magnetic field (B) is subject to a Lorentz force. If a coil of wire length L is placed in a horizontal, radial B field profile, this force is: $F_{\text{Lorentz}} = I L B$

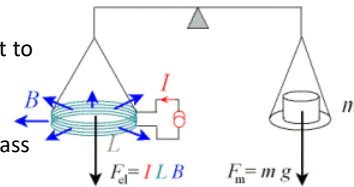
This coil is one side of a sensitive balance, the other side holds the mass to be studied, on which the gravitational force acts: $F_m = m g$.

The coil current is set such that the Lorentz force balances the gravitational force, $F_m = F_{\text{Lorentz}}$.

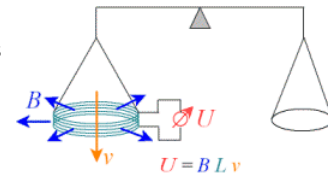
Magnetic field and wire length is difficult to measure with metrological accuracy, so another measurement is also applied. The same coil moves with v velocity in the same magnetic field, which yields an induced voltage of $U = B L v$. Here U and v are measured.

From the two measurements L and B are cancelled yielding, $U_2 I_1 = m g$, where the index 1 denotes the first measurement (balance mode) and index 2 denotes the second measurement (velocity mode). The current measurement is converted to voltage ($I_1 \rightarrow (e^2/h) U_1$, Quantum Hall), while the voltage is converted to frequency ($U_2 \rightarrow (h/2e) f_2$, $U_1 \rightarrow (h/2e) f_1$, Josephson):

Accuracy: $\sim 3.6 \times 10^{-8}$ NIST (2011). Note that g is measured more accurately with a superconducting gravity meter.



Two measurement steps with a Watt balance



$$m = \frac{U_2 I_1}{g v_2}$$

$$U_2 I_1 = U_2 U_1 \frac{e^2}{h} = \frac{h}{2e} f_2 \frac{h}{2e} f_1 \frac{e^2}{h} = \frac{h}{4} f_2 f_1$$

Note the role of Planck's constant!

<https://www.nist.gov/si-redefinition/kilogram-kibble-balance>

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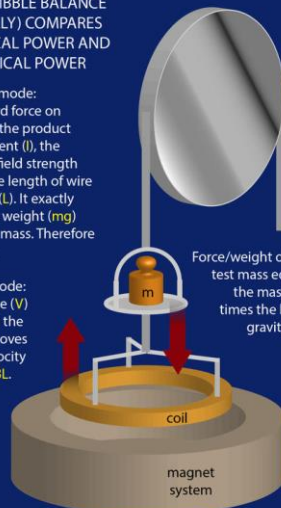
Watt balance: some pictures from the NIST website

HOW A KIBBLE BALANCE (VIRTUALLY) COMPARES ELECTRICAL POWER AND MECHANICAL POWER

Weighing mode:
The upward force on the coil is the product of the current (I), the magnetic field strength (B), and the length of wire in the coil (L). It exactly equals the weight (mg) of the test mass. Therefore $mg = IBL$.

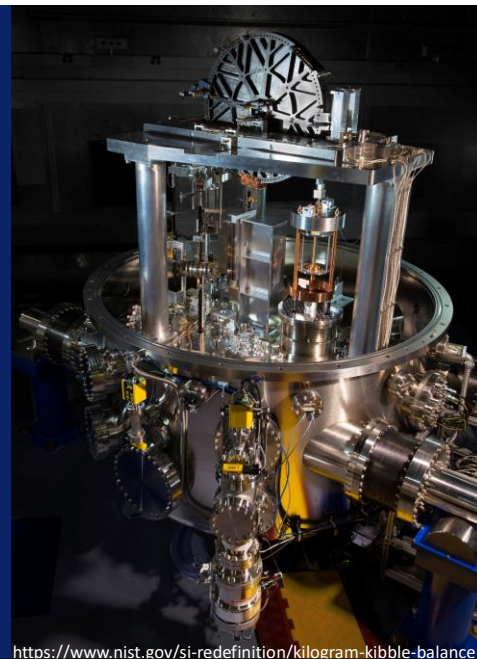
Velocity mode:
The voltage (V) induced in the coil as it moves equals velocity (v) times BL .

Force/weight of the test mass equals the mass (m) times the local gravity (g).



Weighing Mode: $mg = IBL$ Velocity Mode: $V = vBL$
so $mg/I = BL$ so $V/v = BL$

BL is the same in each case and cancels out. Thus
 IV (watts elec. power) = mgv (watts mech. power)



<https://www.nist.gov/si-redefinition/kilogram-kibble-balance>

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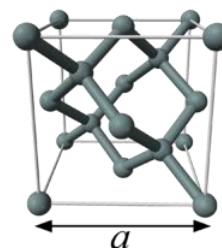
Kilogram and the Avogadro number

Another alternative is to accurately measure the kilogram:
Avogadro project (a large international collaboration).

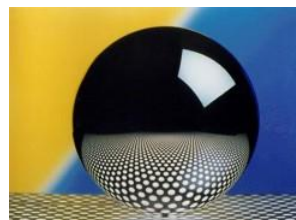
First this project was used to **accurately determine the Avogadro number**. Principle: the number of atoms in a Si sphere of 1kg mass. The lattice constant of Si (a) can be known accurately from X-ray diffraction. From this, the number of atoms in a sphere of a given radius can be determined. Si can be grown in perfect crystal with high purity. The spherical geometry can also be manufactured with high accuracy: $R=9\text{cm}\pm0.3\text{nm}$. Isotope purification is required. Accuracy: 3×10^{-8} (2011).

Once the Avogadro number is fixed, the Avogadro project can be also used as a mass standard. It is not elegant, since Avogadro's number is only a scaling factor between the macro and micro worlds and not a natural constant like h . Also, it is difficult to repeat the measurement.

Crystal structure of Si



Si single crystal sphere



<https://www.nist.gov/si-redefinition/kilogram-silicon-spheres-and-international-avogadro-project>